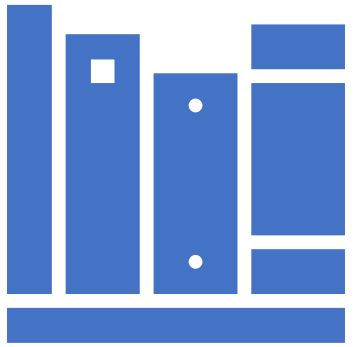






GAILY REY APRIL M. GUZON

CAPSTONE CHAPTERS - GUIDELINES

MANUSCRIPT FORMAT

Based on CCS Capstone Manual and Student Paper Setup Guide, APA Style 7th Edition

Page Size and Margins

- ✓The entire document is Letter 8.5 x 11 inches.
- ✓Portrait orientation
- ✓1-inch margin for top, bottom, and right. 1.5 for the left margin

Type Style

- ✓Font Style – Times New Roman
- ✓Font Size – 12
- ✓Font Size for Tables and Figures – 10

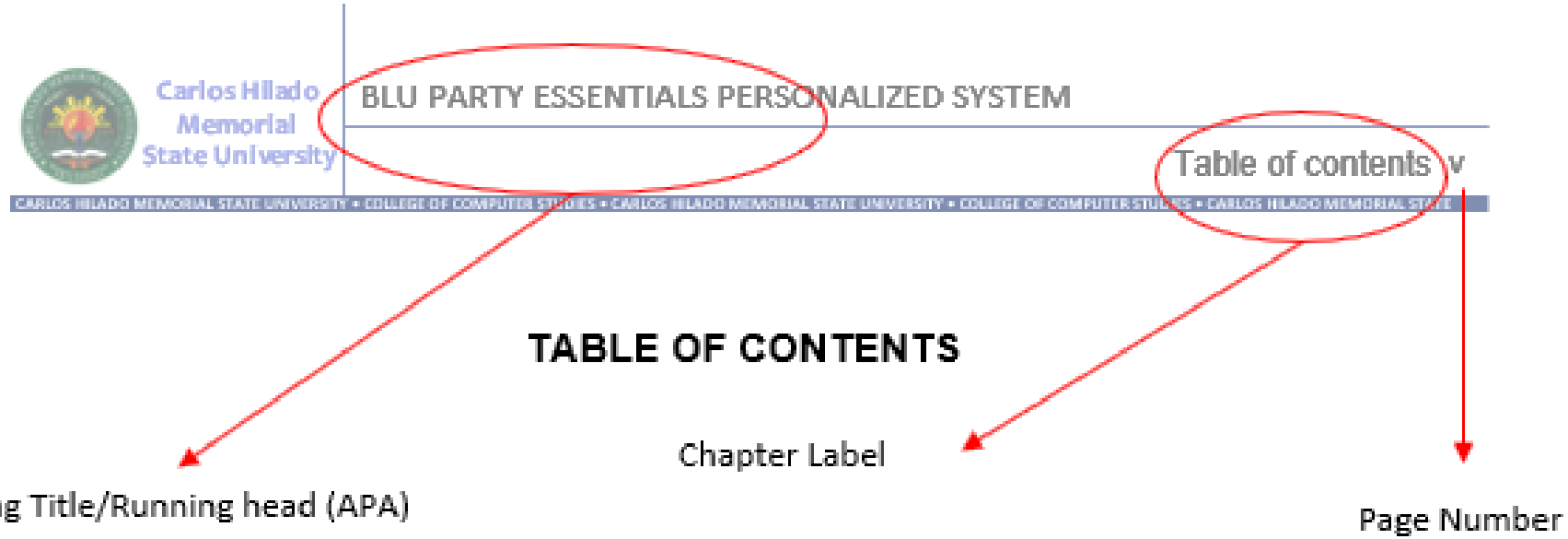
Line Spacing

- ✓Double-spaced (2.0) between all text lines of the manuscript.
- ✓Single space table of contents
- ✓One space between words
- ✓Indent at beginning of each paragraph

Paragraph alignment and indentation

- Align paragraphs of text to **full justification.**
- Indent the first line of every paragraph of text 0.5 inches using the tab key or the paragraph-formatting function of your word-processing program.

Page Heading, Footer, and Pagination



- The running head is a shortened title (no more than 50 characters, including spaces) APA guidelines that appears on every page. Use the automatic functions of your word-processing program to create a header that contains the running head and the page numbers for your paper.

Page Heading, Footer, and Pagination

Page Heading, Footer, and Pagination

- Every page should be assigned a number with the following exceptions, **the title page and the Approval page**, which are counted, but not numbered.
- The Title Page is considered as page ‘i’ and the Approval page ‘ii’.
- Page numbers on these pages using small Roman numerals (i.e., “iii”, “iv”, etc.): Dedication page, Acknowledgements, Abstract, Table of Contents, List of Tables, and List of Figures.
- Beginning with Chapter 1, pages are numbered with Arabic numbers 1 and it continuous throughout the manuscript.
- NOTE: We also omit the chapter labels and page numbers on every introductory page of a Chapter; Chapter 1, Chapter 2, …..

Headings

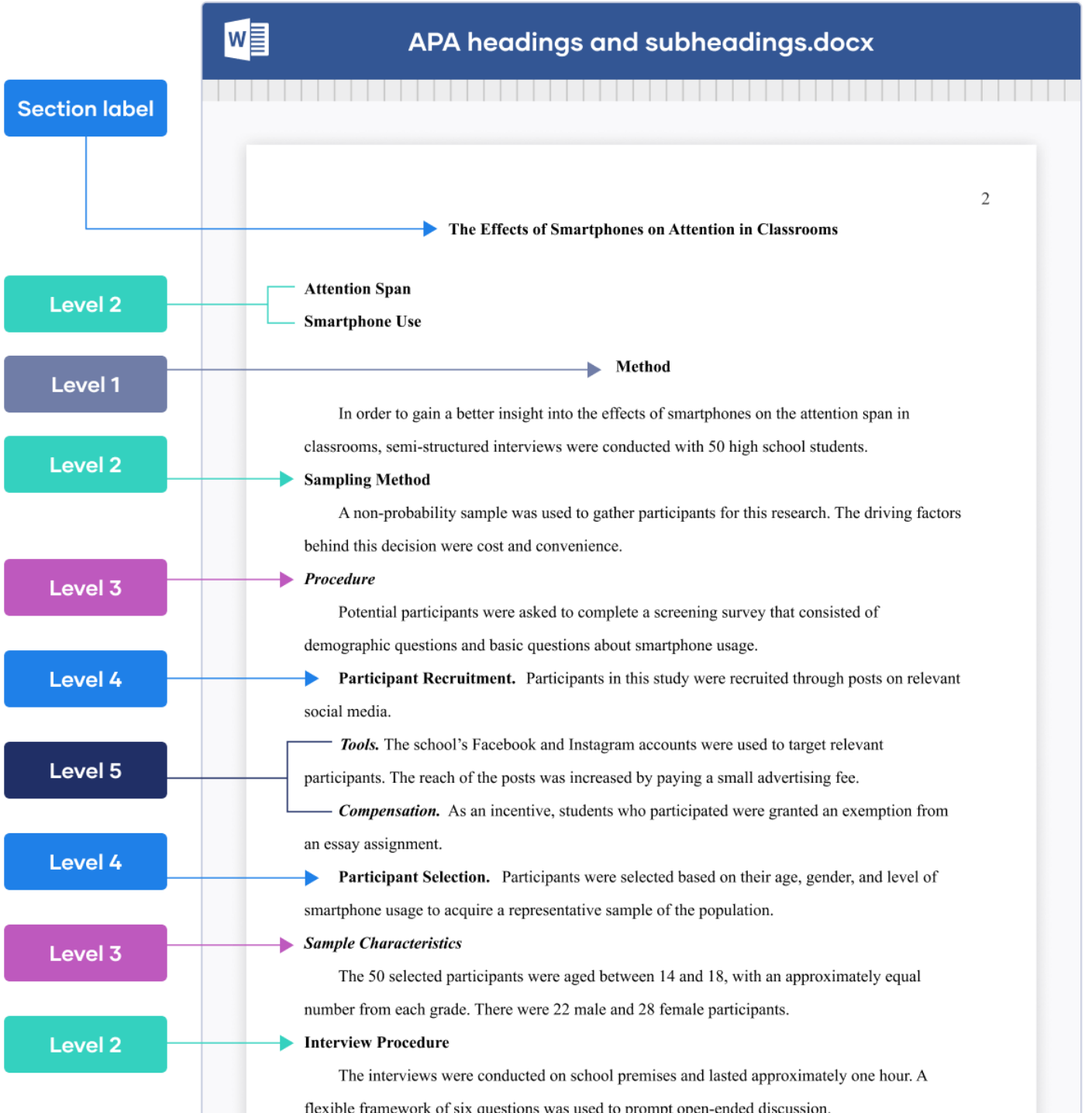
Major Headings

- ✓Sections starting with a Major heading (Heading 1) should begin on a new page.
- ✓Do not place a major heading in the middle of the page.

Subheadings

- ✓Subheadings within a chapter do not begin on a new page unless the preceding page is filled.
- ✓Subheadings at the bottom of a page require at least two lines of text following the heading, otherwise, the heading should be pushed to the next page.

Headings & Subheadings



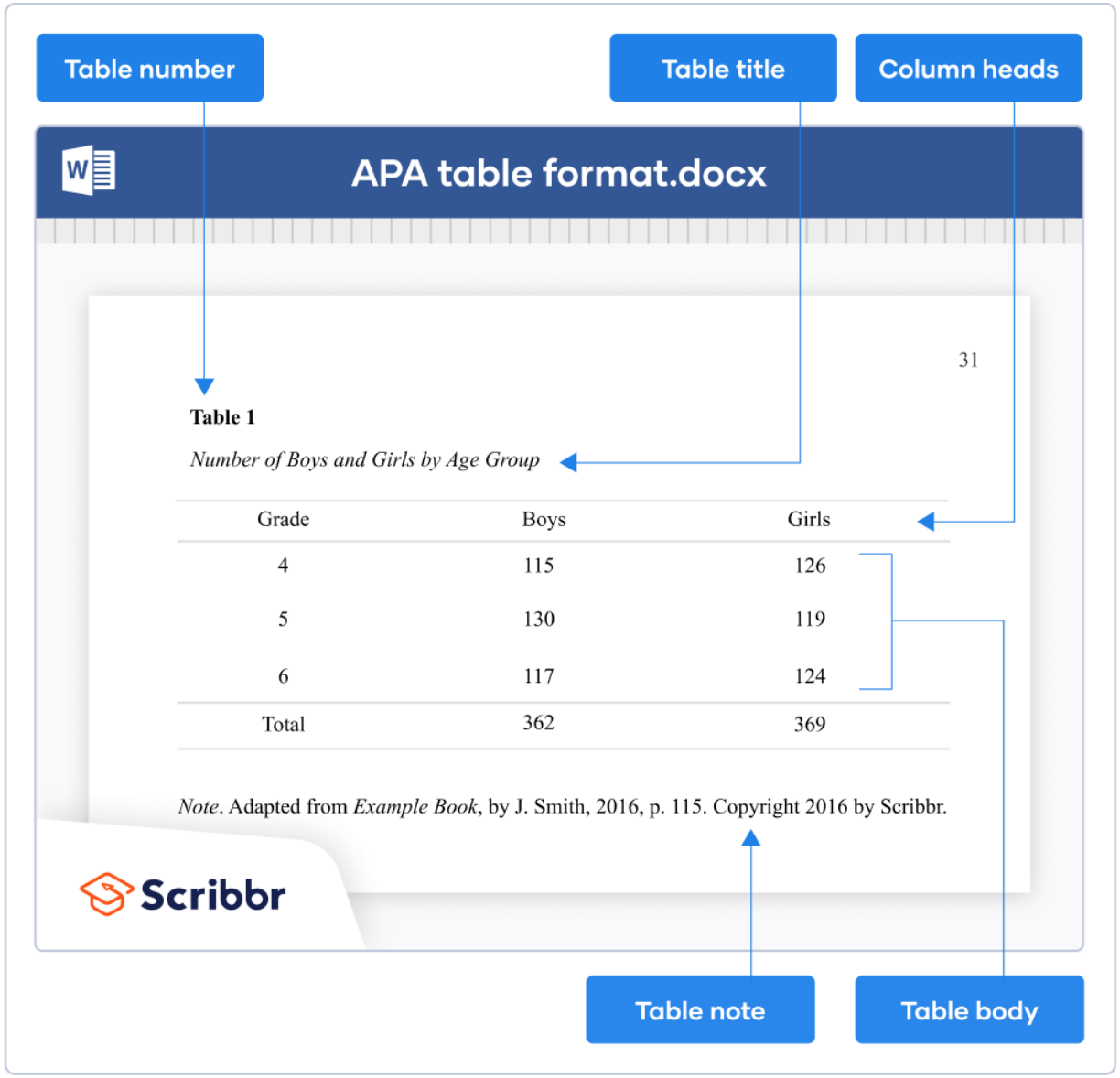
Tables

- Tables and figures must be referenced in your text.
- Figures should use word wrapping that is in line with the text.
- All tables and figures must conform to the margin requirements. Images can be resized to meet the margin requirements OR the page may be landscape orientation for large tables and figures.
- Do not use screenshots of tables from other sources in your work. If you have an image of a table from another source, recreate the table in your word processor.
- All tables, even if there is only one, must be listed in a List of Tables located on separate pages following the Table of Contents.
- The Lists of Tables should include a Table/Figure number along with descriptive text (usually a portion of the caption)

Tables (APA Format)

- Table number in bold above the table.
- Brief title, in italics and title case, below the table number.
- No vertical lines.
- Horizontal lines only where necessary for clarity.
- Clear, concise labels for column and row headings.
- Numbers consistently formatted (e.g. with the same number of decimal places).
- Any relevant notes below the table.

Tables (APA Format)



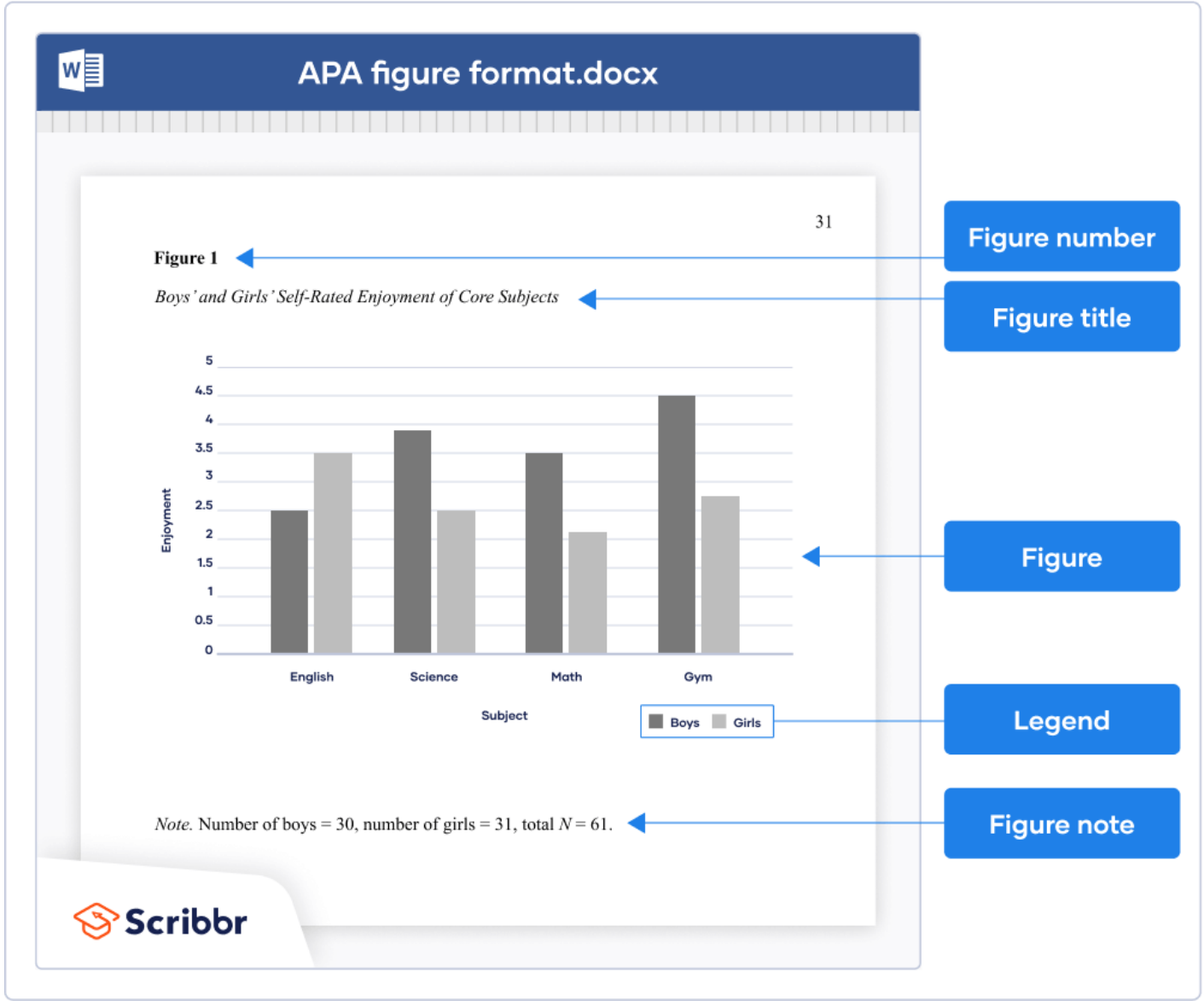
Figures

- Figures may not appear on the title page or other introductory pages.
- Diagrams, drawings, graphs, charts, maps, photographs, or any other type of illustrations should be referred to in the manuscript as a ***Figure***.
- Figures must be referenced in your text.
- Figures should use word wrapping that is in line with the text.
- All figures, even if there is only one, must be listed in a List of Figures located on separate pages following the Table of Contents.
- The Lists of Figures should include a Figure number along with descriptive text (usually a portion of the caption)

Figures (APA Format)

- Figure number in bold above the figure.
- Brief title, in italics and title case, under the figure number.
- If necessary, clear labels and legends integrated into the image.
- Any relevant notes below the figure.

Figures (APA Format)



Large Tables and Figures

- Large tables or figures can be placed on the page in landscape orientation
- The top and bottom margins should be the same as on a regular page

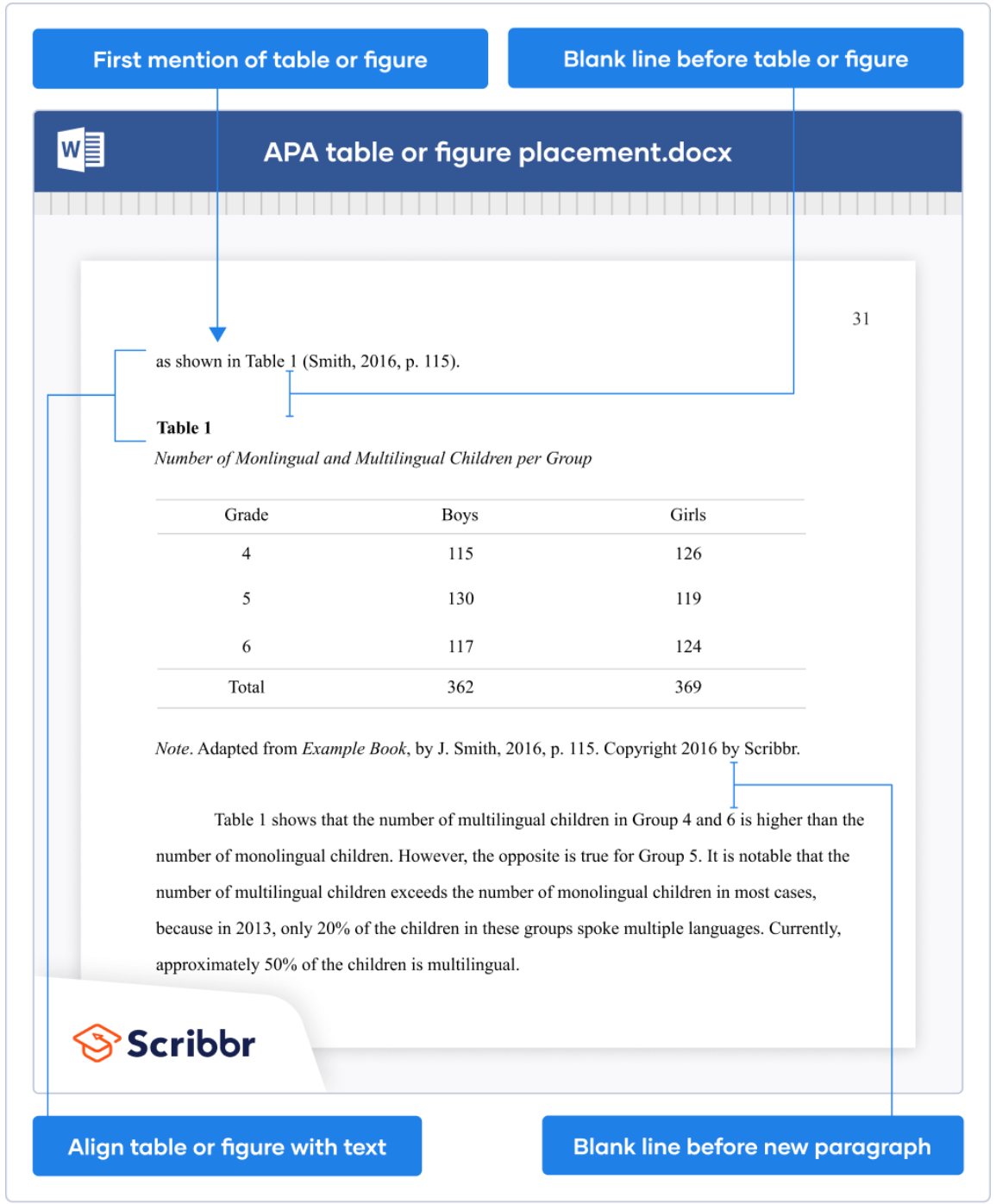
Captions and Numbering

- ✓All tables and figures in your paper must include a proper caption located either above or below the table or figure. Be consistent in placement
- ✓Tables and figures should be numbered and ordered sequentially as they appear in your paper
- ✓Tables and figures in appendices should be captioned and continue the numbering sequence used in the text

Where to place tables and figures

- You have two options for the placement of tables and figures in APA Style:
- **Option 1:** Place tables and figures throughout your text, shortly after the parts of the text that refer to them.
- **Option 2:** Place them all together at the end of your text (after the reference list) to avoid breaking up the text.
- If you place them throughout the text, note that each table or figure should only appear once. If you refer to the same table or figure more than once, don’t reproduce it each time—just place it after the paragraph in which it’s first discussed.
- Align the table or figure with the text along the left margin. Leave a line break before and after the table or figure to clearly distinguish it from the main text, and place it on a new page if necessary to avoid splitting it across multiple pages.

Where to place tables and figures



References

- ✓Use APA 7th Edition
- ✓Arrange alphabetically
- ✓Make sure that everything you cite in the text is also in the reference list and vice versa
- ✓Apply a hanging indent of 0.5 inches.

APA 7th Edition References

FORMAT

Primary contributors (Publication date). *Title* (Edition). Publisher.

Parenthetical (Primary contributors, Publication date)

APA reference entry

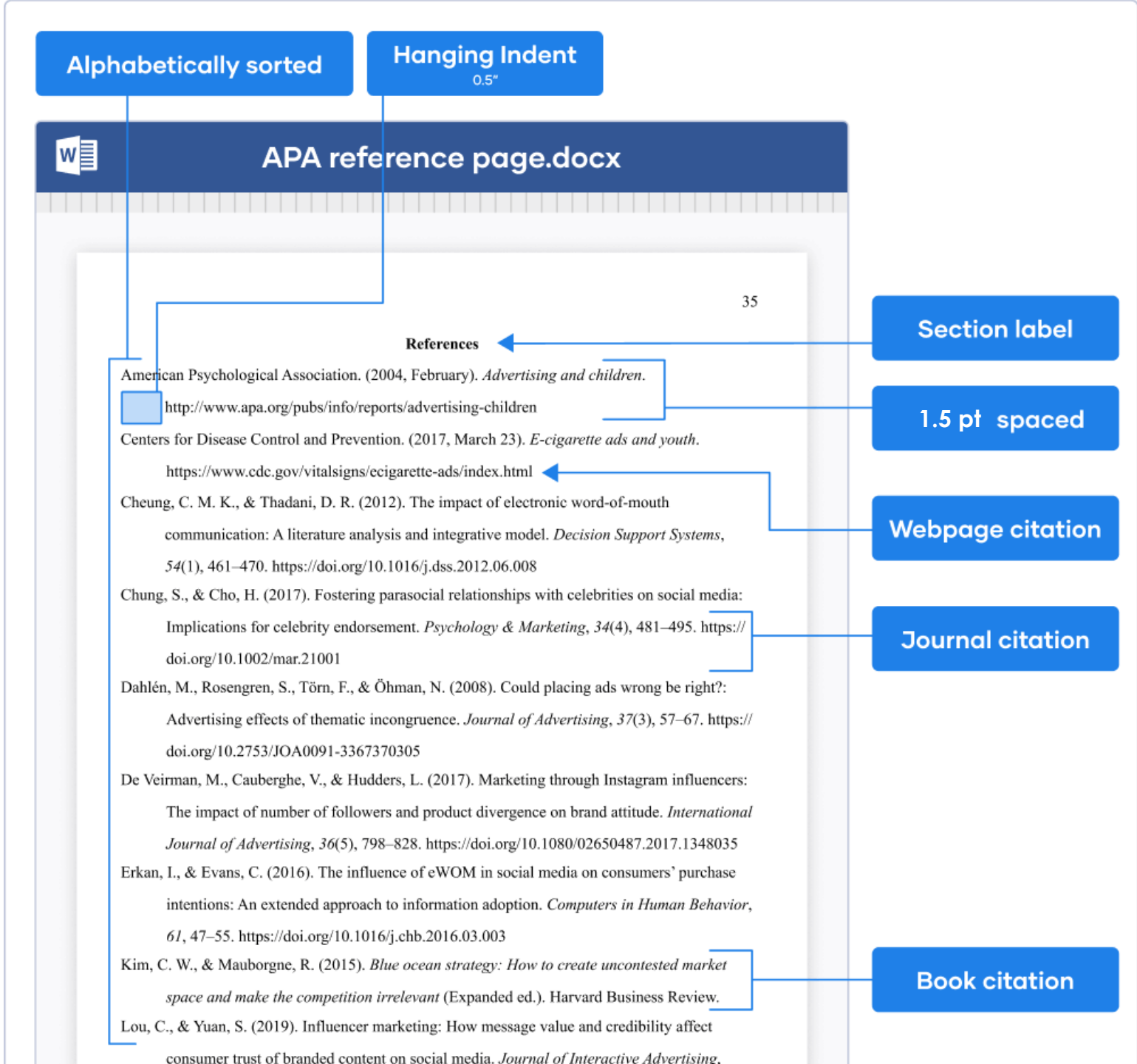
APA in-text citation

EXAMPLE

Smith, T. (2020). *The citation manual for students: A quick guide* (2nd ed.). Wiley.

Parenthetical (Smith, 2020)

References


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References
American Psychological Association. (2004, February). *Advertising and children*. http://www.apa.org/pubs/info/reports/advertising-children
Centers for Disease Control and Prevention. (2017, March 23). *E-cigarette ads and youth*. https://www.cdc.gov/vitalsigns/ecigarette-ads/index.html
Cheung, C. M. K., & Thadani, D. R. (2012). The impact of electronic word-of-mouth communication: A literature analysis and integrative model. *Decision Support Systems*, *54*(1), 461–470. https://doi.org/10.1016/j.dss.2012.06.008
Chung, S., & Cho, H. (2017). Fostering parasocial relationships with celebrities on social media: Implications for celebrity endorsement. *Psychology & Marketing*, *34*(4), 481–495. https://doi.org/10.1002/mar.21001
Dahlén, M., Rosengren, S., Törn, F., & Öhman, N. (2008). Could placing ads wrong be right?: Advertising effects of thematic incongruence. *Journal of Advertising*, *37*(3), 57–67. https://doi.org/10.2753/JOA0091-3367370305
De Veirman, M., Cauberghe, V., & Hudders, L. (2017). Marketing through Instagram influencers: The impact of number of followers and product divergence on brand attitude. *International Journal of Advertising*, *36*(5), 798–828. https://doi.org/10.1080/02650487.2017.1348035
Erkan, I., & Evans, C. (2016). The influence of eWOM in social media on consumers' purchase intentions: An extended approach to information adoption. *Computers in Human Behavior*, *61*, 47–55. https://doi.org/10.1016/j.chb.2016.03.003
Kim, C. W., & Mauborgne, R. (2015). *Blue ocean strategy: How to create uncontested market space and make the competition irrelevant* (Expanded ed.). Harvard Business Review.
Lou, C., & Yuan, S. (2019). Influencer marketing: How message value and credibility affect consumer trust of branded content on social media. *Journal of Interactive Advertising*,
Section label
1.5 pt spaced
Webpage citation
Journal citation
Book citation

CHAPTER 1

INTRODUCTION

Background of the Study

The Research Description/Project Context

- The researcher must describe the chosen problem through the background of the study. This includes the description of the client, the existing system and its prevailing issues based on his data gathering.
- Project context gives a strong justification for choosing such research problem or rationale.
- Finally, in the project context, provide a statement that shows the relationship among the rationale of the study to the proposed research problem.
- Why is there a need of developing such system?

Background of the Study

Purpose and description

- A **purpose** statement is a declarative sentence which summarizes the specific topic and goals of a document. It is typically included in the introduction to give the reader an accurate, concrete understanding what the document will cover and what he/she can gain from reading it.

- NOTE: All sources must be cited and included in the References

Objectives of the Study

- A research objective is a concrete statement that describes what the researcher is trying to achieve. A well-worded objective will be SMART, i.e.
- **Specific**
- **Measurable**
- **Attainable**
- **Realistic, and**
- **Time-bound**.

Objectives of the Study

General Objective (Level 2 Heading)

- General objectives are extensive goals to be achieved and are usually less in number.
  - The study aims to design and develop the __”Title”____ for the _____.
- **Specific Objectives (Level 3 Heading)**
- Specific objectives are short term and constricted in focus.
- General objectives can be broken into smaller parts to form specific objectives.
- Most of all, specific objectives gives a clearer vision of what the study is trying to achieve.

Objectives of the Study

- **Specific Objectives (Level 3 Heading)**
- Every statement must start with “To…”
- Example:
  - To design the system with following features:
    a.
    b.
    c.
  - To evaluate the quality of the system.
    - Experts evaluation (5 members)
  - To evaluate the acceptability of the system.
    - End-users evaluation (10+ members)
  - To develop User Manual

Scope and Limitations of the Study

- The scope explains the nature, coverage and time frame of the study. 1st paragraph
  - May explain the scope of the study including the functionalities and features of the system, the users and level of users of the system, the expected outputs of the system, etc.
- The limitations explains all that are **NOT** included in your project, usually in the 2nd paragraph.

Significance of the Study

- Describes the contributions of the study as new knowledge and makes findings more conclusive.
- Also, it cites the usefulness of the study to users and certain groups.
- Then, cite all the persons and groups that would benefit from the study.
- Users from the Use Cases must also be included.

Definition of Terms

- Definition of terms gives definition to the major terms that are relevant to your study for the clarification of the common understanding.
- Lastly, the definition can either be conceptual or operational or both in relation to the study only.
- Should be in **chronological order.**

- NOTE: All sources must be cited and included in the References

Conceptual Framework

- A conceptual framework illustrates the expected relationship between your variables. It defines the **relevant objectives** for your **research process** and maps out how they come together to draw coherent conclusions.

Conceptual Framework

IPOO Framework

- The input-process-output-outcome model is a conceptual framework used in software development in the field of information technology.
- This model involves four main components: input, process, output , outcome and output.
- The **input** refers to the data or information that is provided to the system for processing.
- The **process** involves the actions or operations performed on the input to transform it into meaningful output -> system.
- The **output** is the product of the processing that is generated.
- The **outcome** is the benefit or impact of the product to its target users.
- This model helps in understanding the flow of data and the steps involved in the development of software systems.

IPO Model Conceptual Framework of Voting System

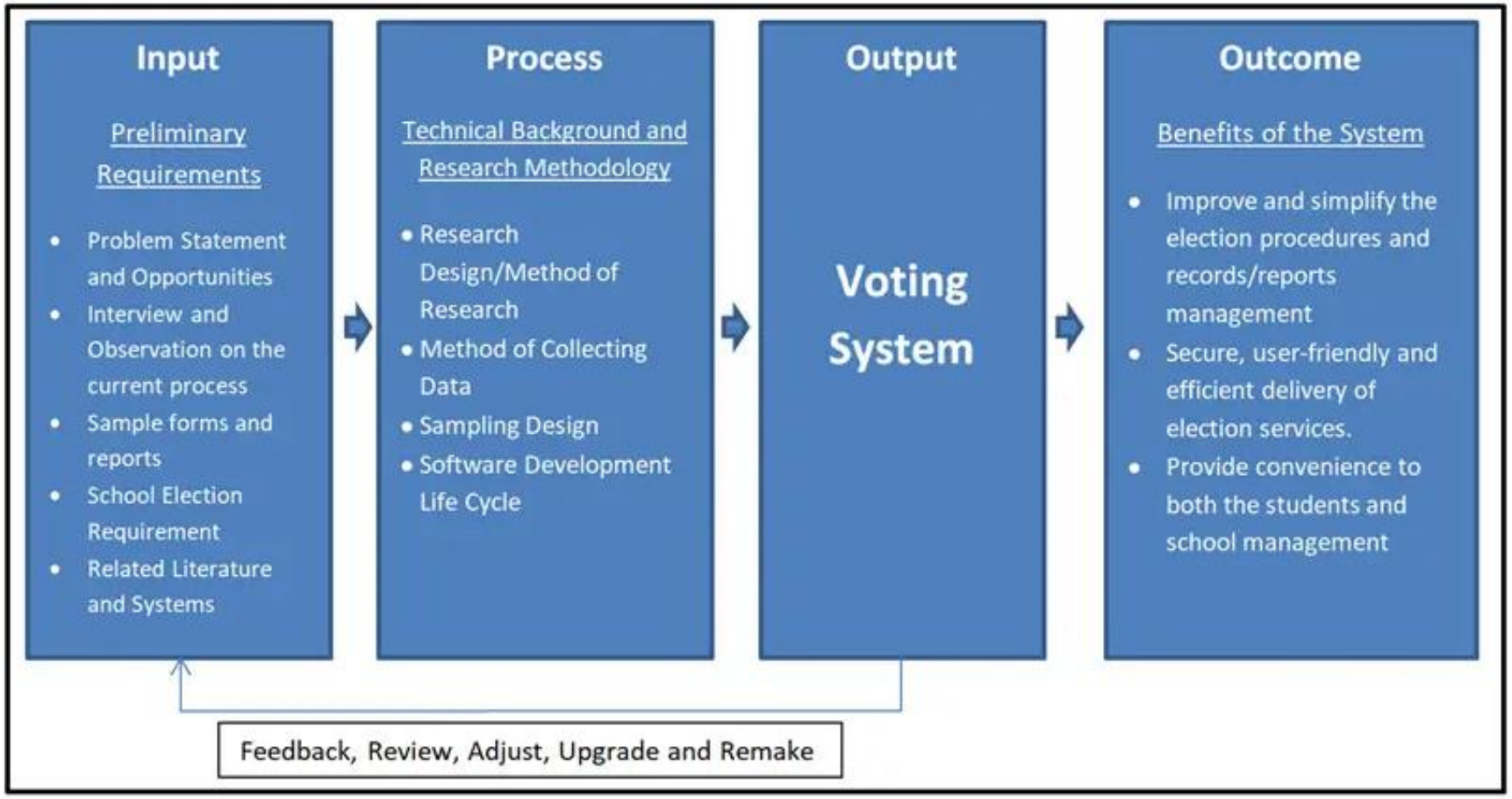


FIGURE LABEL IN MANUSCRIPT

Figure 1

Conceptual Framework of _____





CHAPTER 2

REVIEW OF RELATED LITERATURE

REVIEW OF RELATED LITERATURE

- In this chapter, you must start with a brief introduction about the investigations of the researcher on the review literature and system about the problem.

- Here, you must get your data from any book, magazines, researchers and much more. Most of all, label your published material with local or foreign sources.

Things to consider in writing chapter 2:

- First, include only those references that have immediate connections or relevance for the study and to the system.
- Second, be explicit in stating the relevance of all cited literature and system.
- Third, the review should be in the form of an integrated presentation of all materials.
- Next, it must not contain a simple listing (annotated bibliography/citation/reference) in APA 7th edition format
- Also, when comparing and summarizing studies, be sure to note possible differences between samples, measurement techniques and much more, summarize only those truly comparable sources.

Things to consider in writing chapter 2:

- Then, it is best that your sources is updated and is finished in the most recent years. Probably, more effective if the material should not be older than 10 years
- Another, to construct a literature review consider the following issues:
  - the **selection of the problem**
  - **delimiting** the **scope of the problem**
  - demonstrating the relationship of the study to **previous knowledge**
  - indicate the **contribution of the study**, and
  - examine the **tools** and methods in **investigating** the problem.
- In addition, the researcher must be selective with respect to the **quality of the sources cited.**

Sources

Foreign Related Literature (3)

- The foreign related studies are the studies conducted internationally or outside the Philippines.

Local Related Literature (3)

- The local related studies are the studies conducted here locally or within the Philippines only.

Foreign Prior Art Systems (3)

- The foreign related systems are the systems designed and developed internationally or outside the Philippines.

Local Prior Art Systems (3)

- The local related studies are the systems designed and developed locally or within the Philippines only.

Feature Comparison of Local and Foreign Related Literature

The table of comparison shows the different local and foreign prior art systems that were compared to the “Title”. This table compares the prior art systems and its features to the proponents’ proposed system.

Feature Comparison of Local and Foreign Related Literature

The table of comparison shows the different local and foreign prior art systems that were compared to the “Title”. This table compares the prior art systems and its features to the proponents’ proposed system.

Table 1

Table of Comparison of Local and Foreign Prior Art Systems

PRIOR ART SYSTEM	FEATURES		
	FEATURE 1	FEATURE 2	FEATURE 3
SYSTEM 1	YES/NO	YES/NO	YES/NO
SYSTEM 2	YES/NO	YES/NO	YES/NO
SYSTEM 3	YES/NO	YES/NO	YES/NO
SYSTEM 4	YES/NO	YES/NO	YES/NO
SYSTEM 5	YES/NO	YES/NO	YES/NO
TITLE	YES/NO	YES/NO	YES/NO

CHAPTER 3

DESIGN AND METHODOLOGY

Capstone Research Components

- **Research Design**
  - Brief discussion of what research design will be used in the study
- **Respondents of the Study**
  - Target users of the proposed system or Community where the research will be conducted and where research participants will be drawn
- **Sample Size**
  - When you conduct research about a group of people, it's rarely possible to collect data from every person in that group. Instead, you select a sample. The sample is the group of individuals who will actually participate in the research.
  - The number of respondents or participants of the study
- **Sampling Method**
  - To draw valid conclusions from your results, you have to carefully decide **how you will select a sample that is representative of the group as a whole.** This is called a **sampling method.** There are two primary types of sampling methods that you can use in your research:
    - **Probability sampling** involves random selection, allowing you to make strong statistical inferences about the whole group.
    - **Non-probability sampling** involves non-random selection based on convenience or other criteria, allowing you to easily collect data.
  - You should clearly explain how you selected your sample in the methodology section of your paper or thesis, as well as how you approached minimizing research bias in your work.

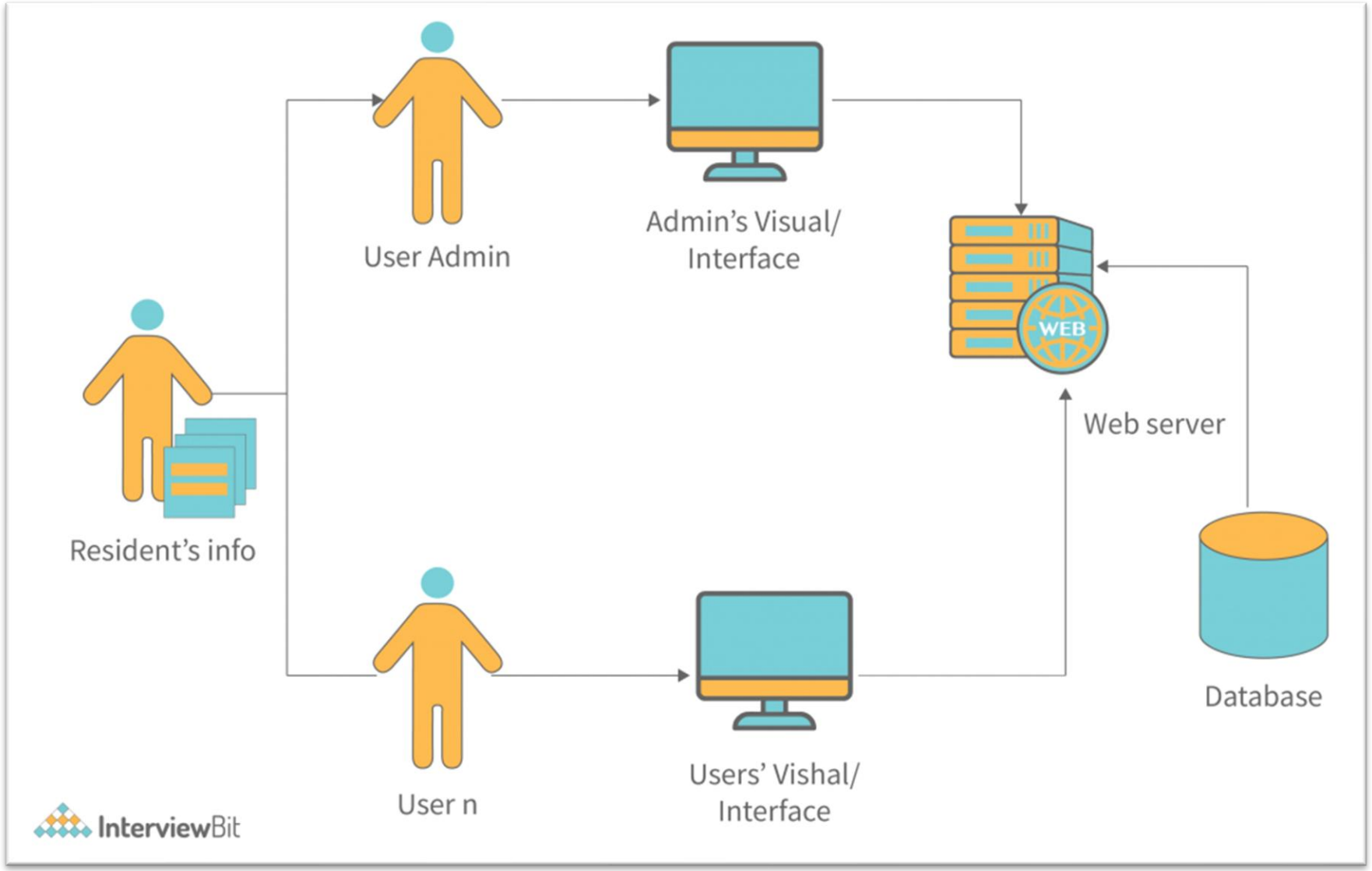
Capstone Research Components

- **Data Gathering Instrument**
  - Tools/Materials/Questionnaires to be used in the study as an evaluation tool
- **Data Gathering Procedure**
  - Method of collecting data and the development of instrument for gathering data must be explained.
  - How the research will be conducted, particularly the evaluation of the proposed system
- **Statistical Treatment of Data**
  - Statistical treatment of data is a process used to convert raw data into something interpretable.
  - This is done right after the data gathering.
- **Ethical Considerations**
  - Ethical considerations in research are a set of principles that guide your research designs and practices. These principles include voluntary participation, informed consent, anonymity, confidentiality, potential for harm, and results communication.

Capstone Developmental Components Research

- **System Development Method**
  - Choose an SDLC that you will be using as a methodology for the development of the System (ex. Agile or RAD (Rapid Application Development))
  - Explain your activities during each phases (ex. planning)
- **System Architecture**
  - Software architecture is a set of principles that define the way software is designed and developed.
  - The system architecture diagram is a visual representation of the system architecture. It shows the connections between the various components of the system and indicates what functions each component performs. The general system representation shows the major functions of the system and the relationships between the various system components

System Architecture Example



Capstone Developmental Components Research

- **Use Case Diagram**
- **Use Case Description**
- **Context Diagram**
- **Data Flow Diagram**
  - Level 1 and Level 2
- **Entity Relationship Diagram**
- **Data Dictionary**
- **Hardware and Software Requirements**
- **Gantt Chart**

CHAPTER 4

RESULTS AND DISCUSSION

Overview of Chapter 4

- This chapter presents the findings of the research.
- The data collected from testing, analysis, and feedback are presented in various forms such as tables, charts, or graphs.
- The results are then discussed, relating them to the objectives of the study and comparing them with the findings from previous research.

PRESENTATION OF RESULTS

Presentation of Results

- Present data related to the system’s performance, such as speed, efficiency, accuracy, and other parameters in Table.
- When presenting quantitative results, such as mean scores and their corresponding verbal interpretations, it is important to follow a consistent and clear format for readability and accuracy.
- Ensure that all mean scores in the table are rounded to the same number of decimal places, typically **two** (e.g., 4.56).
- Each mean score should have a corresponding **verbal interpretation** based on the defined criteria in the study (e.g., 4.5 – 5.0 = "Excellent", 3.5 – 4.4 = "Very Good"). This helps in providing a qualitative understanding of the quantitative data.

Presentation of Results

- Example:

Table 23

Results of User Evaluation

Criteria	Mean Score	Verbal Interpretation
User Interface Usability	4.67	Excellent
System Performance	3.92	Very Good
Response Time	4.20	Very Good
User Satisfaction	4.78	Excellent
Error Rate	2.65	Fair

Interpreting Mean Scores

- Ensure that a predefined **range** for each verbal interpretation is clearly defined in the study (Chapter 3), and must be consistent with the results in Chapter 4 and 5.
- Example:

4.21 – 5.00 = Excellent

3.41 – 4.20 = Very Good

2.61 – 3.40 = Fair

1.81 – 2.60 = Poor

1.00 – 1.80 = Very Poor

User Testing Results

- Show the results from user testing, feedback, or surveys. Highlight usability, user satisfaction, or any identified parameters on your objectives. Use quantitative data where possible.
- Example:

User testing was conducted with [number] participants, focusing on system usability and functionality. The majority of users rated the system [data, e.g., 4.40 out of 5.00] in terms of ease of use. However, some participants noted issues with [specific part], suggesting that [mention specific improvements].

System Feature Testing

- Present results related to specific system features, outlining their success or failure in meeting the project's objectives.

Example:

The feature for [specific feature] performed as expected, handling [mention volume or type of data] without issues. However, during testing, [mention any limitations], which suggests the need for [suggest improvements].

DISCUSSION OF RESULTS

Interpretation of Results

- Interpret the results in light of the research objectives. Discuss whether the results met expectations and why.

Example:

The results indicate that the system met the project's objectives for [mention objective]. The performance of [specific feature] exceeded initial estimates, which can be attributed to [mention reason]. However, [specific area] fell short due to [mention any issues or challenges encountered].

CHAPTER 5

SUMMARY OF FINDINGS, CONCLUSION, AND RECOMMENDATIONS

Summary of Findings

- Summarize the key findings from the research, referring to the results from Chapter 4.

Example:

- *The study found that the system successfully met the primary objective of [mention primary objective]. It performed well in terms of [specific metrics], achieving [mention key results]. However, [specific limitation] was identified during testing, which affected [specific aspect]. Overall, user feedback indicated high satisfaction with [mention feature].*

Conclusion

- Provide a conclusion based on the findings. Relate this conclusion back to the research problem and objectives.

- Example:

In conclusion, the system developed in this study has proven to be an effective solution for [research problem]. It met most of the objectives, particularly in terms of [mention key success]. However, some aspects, such as [mention limitation], require further refinement. This project has shown that [mention broader significance of the findings].

Recommendations

- Offer recommendations for future work, system improvement, or further research based on the findings

Recommendations given by the Panel Members

- Consider the suggestions and recommendation for improvement of the members of the Panel during the Final Presentation of your project.

System Improvements

- Recommend ways to improve the system based on identified issues or limitations.

- Example:

To improve the system, it is recommended to enhance [specific feature], as user feedback indicated that [mention issue]. Additionally, incorporating [mention specific tool or method] could further increase the system's performance in [mention area].

Future Research

- Suggest areas for future research that could build on the findings of the study.

- Example:

Future research could explore [mention potential topic], as this study was limited in scope in that area. Additionally, research into [mention related technology or feature] could provide insights into improving [specific aspect].

Implementation

- Discuss how the system or findings could be applied in other contexts, industries, or utilized by the users.

- Example:

Given the success of the system in [specific domain], it could potentially be adapted for use in [mention other industry or context]. Further exploration of this idea could reveal opportunities for broader applications.